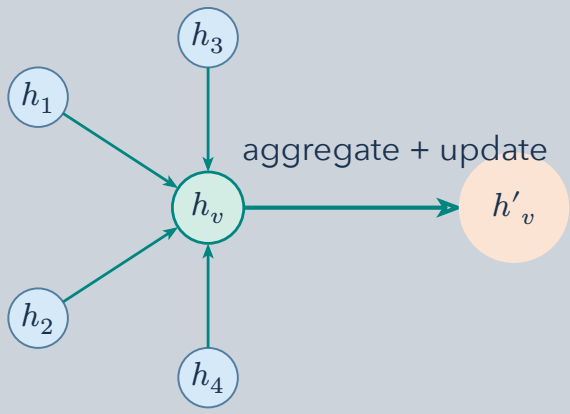
A node learns from its neighborhood. These views connect the local operation, the growth of its receptive field, and two ways to construct the information being combined.

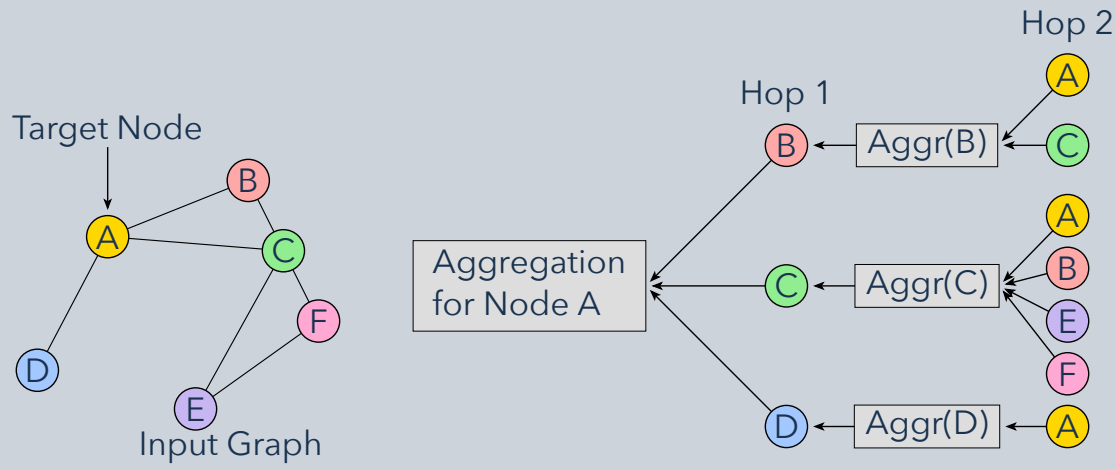
1 Aggregate the neighbors



sum or mean ignores neighbor ordering

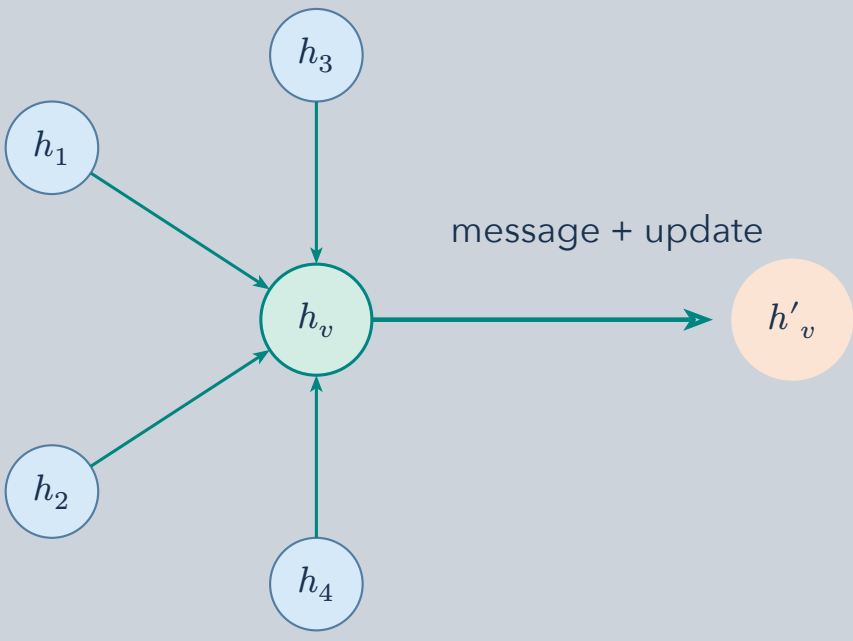
Graph convolution mixes neighboring node features with shared parameters. A sum or mean is unchanged when the neighbors are reordered; include self-information in the update.

2 Stack layers to reach farther



One layer communicates across one edge; two layers can use two-hop information. The expanded tree shows computation paths, not duplicated physical nodes.

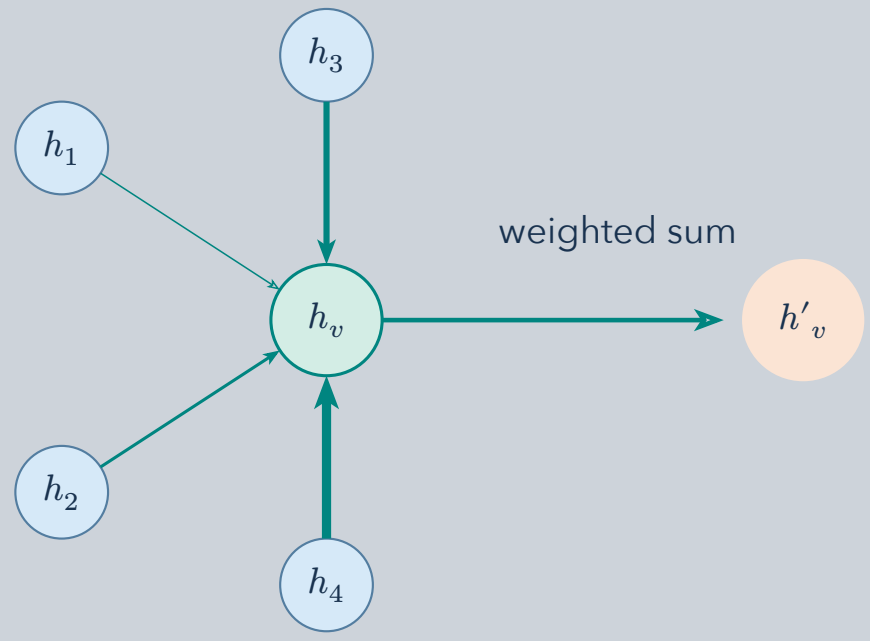
3 Construct messages



$$m_{uv} = M(h_u, h_v, e_{uv})$$

A message function M can depend on sender features, receiver features, and edge attributes e_{uv}. Aggregate the messages, then update the receiving node.

4 Learn which neighbors matter



thicker arrow = larger learned weight

Attention assigns normalized weights to neighbors before aggregation. Multiple heads learn different weightings, then concatenate or average their outputs.

h_v = features at node v; h'_v = updated features. **Shared local rules preserve node relabeling symmetry.** A graph-level prediction can pool node features into one permutation-invariant representation.